

Theorem 9.3.3 The Inclusion/Exclusion Rule for Two or Three Sets

If A , B , and C are any finite sets, then

$$N(A \cup B) = N(A) + N(B) - N(A \cap B)$$

and

$$N(A \cup B \cup C) = N(A) + N(B) + N(C) - N(A \cap B) - N(A \cap C) \\ - N(B \cap C) + N(A \cap B \cap C).$$